

BORYEONG, Korea, Republic ofWMO: **471500**Lat: **36.3272N**Lon: **126.5575E**Elev: **55**StdP: **14.67**Time Zone: **9.00 (E09)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.2	18.9	3.1	6.5	21.4	6.2	7.6	23.2	12.6	37.6	11.7	36.4	4.8	20	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.7	89.3	79.9	86.8	78.8	84.5	77.4	82.1	86.2	80.7	84.2	79.2	82.7	4.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.1	162.0	84.5	79.7	154.3	83.1	78.3	147.0	81.9	46.1	87.0	44.5	84.3	42.9	82.8	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.7	12.8	10.8	DB	13.4	92.7	4.4	2.4	10.2	94.4	7.6	95.8	5.1	97.1	1.9	98.9	(4)
(5)			WB	11.6	82.2	3.8	2.7	8.9	84.2	6.6	85.8	4.5	87.3	1.7	89.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	30.8	34.7	42.1	51.7	62.3	70.7	77.4	79.2	71.0	60.2	48.3	35.4		
	DBStd	17.47	6.66	6.71	6.18	6.03	5.06	3.30	3.60	4.12	4.05	5.51	7.50	7.28		
	HDD50	1922	597	431	261	53	0	0	0	0	4	122	454			
	HDD65	4728	1062	849	711	401	113	3	0	0	4	166	502	917		
	CDD50	3896	0	2	15	103	381	621	849	905	629	319	70	2		
	CDD65	1227	0	0	0	1	29	174	384	440	183	15	1	0		
	CDH74	8663	0	0	0	8	111	623	2930	4143	810	38	0	0		
	CDH80	2067	0	0	0	0	5	44	627	1295	93	3	0	0		
	WSAvg	4.0	3.6	3.8	4.2	4.3	4.3	3.5	5.1	4.0	3.4	3.4	3.9	4.1		
	PrecAvg	47.6	1.1	1.3	1.9	3.4	3.5	5.3	10.4	10.0	5.2	1.9	2.3	1.3		
(15) Precipitation	PrecMax	74.7	2.8	3.5	3.8	7.3	9.4	12.9	24.8	25.9	11.1	4.8	6.5	2.7		
	PrecMin	28.5	0.2	0.1	0.1	0.6	0.4	0.2	3.2	3.7	0.2	0.2	0.5	0.3		
	PrecStd	12.0	0.8	1.0	1.0	1.9	2.1	3.6	5.4	5.3	3.5	1.2	1.5	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	53.3	62.5	65.3	73.5	80.7	83.5	90.9	93.4	85.5	78.3	69.9	59.1			
	0.4% MCWB	47.1	55.9	57.1	57.0	63.1	69.2	78.9	80.5	73.4	66.7	61.2	52.3			
	2% DB	47.5	53.9	60.5	68.0	77.0	81.2	87.7	90.6	82.5	74.2	65.9	52.8			
	2% MCWB	43.1	49.4	53.9	56.6	63.5	69.4	79.9	80.3	72.3	65.3	59.1	47.5			
	5% DB	43.7	48.4	56.9	65.3	74.5	79.2	85.4	88.5	80.4	71.8	62.3	49.3			
	5% MCWB	39.1	42.3	50.3	55.0	62.4	69.1	78.7	79.5	71.6	62.9	56.2	44.7			
	10% DB	41.0	45.0	53.1	62.6	72.0	77.6	83.4	86.4	78.5	69.5	59.7	46.2			
	10% MCWB	36.7	39.5	46.6	53.7	61.3	68.9	77.7	78.7	70.3	61.1	54.4	41.5			
	0.4% WB	47.9	56.7	58.9	63.6	71.9	75.6	83.6	84.8	77.8	71.6	63.5	54.3			
	0.4% MCDB	51.3	61.4	62.7	66.5	73.5	78.5	86.7	88.9	82.4	76.5	68.2	57.4			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	43.6	49.4	55.7	59.8	66.7	73.9	81.8	82.8	75.5	68.3	60.9	48.7			
	2% MCDB	47.4	53.4	59.0	65.2	71.4	76.9	85.0	87.5	79.8	72.3	63.9	52.0			
	5% WB	40.0	43.5	51.7	57.7	64.7	72.3	80.6	81.3	74.1	65.7	57.8	45.0			
	5% MCDB	43.1	47.2	56.2	63.0	70.8	75.9	83.3	85.7	78.1	68.8	61.1	48.2			
	10% WB	37.1	40.2	47.6	55.4	63.1	70.7	79.1	79.9	72.7	63.0	55.2	42.0			
	10% MCDB	40.3	44.2	51.9	60.6	69.2	74.9	81.8	84.0	76.7	66.8	58.9	45.9			
(35) Mean Daily Temperature Range	MDBR	12.8	13.8	15.3	16.3	15.4	12.6	9.2	10.7	13.2	15.5	13.4	12.4			
	5% DB MCDBR	14.7	18.2	18.8	19.7	18.5	14.8	11.6	13.4	15.0	16.8	14.5	14.7			
	5% DB MCWBR	11.2	13.8	13.2	11.2	8.9	6.6	5.8	5.9	7.2	9.1	9.9	12.0			
	5% WB MCDBR	13.9	17.1	16.4	16.2	15.0	10.9	9.9	11.8	11.2	13.2	12.7	13.7			
	5% WB MCWBR	11.7	14.4	13.0	10.6	9.5	6.7	6.4	6.3	6.3	9.6	10.6	12.6			
(40) Clear-Sky Solar Irradiance	taub	0.457	0.499	0.577	0.602	0.622	0.694	0.646	0.581	0.510	0.497	0.481	0.457			
	taud	1.911	1.844	1.703	1.694	1.706	1.632	1.783	1.940	2.055	1.952	1.937	1.928			
	Ebn at Noon	217	225	220	225	223	207	216	228	237	224	208	207			
	Edh at Noon	47	56	71	75	75	81	69	58	49	50	45	43			
(44) All-Sky Solar Radiation	RadAvg	752	1023	1395	1623	1787	1642	1397	1455	1337	1162	784	656			
	RadStd	52	92	137	126	179	164	204	214	137	91	88	50			

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	+85	(47)

Nomenclature: See separate page